COMPANION FOR CHAPTER 1

OF

FUNDAMENTALS OF PROBABILITY

WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 1

Axioms of Probability

1A ADDITIONAL EXAMPLES

Example 1a In a mathematics department of 46 voting faculty members, there are two candidates, Ryan and Nolan, running for the chair position. Assuming that each of the 46 faculty members votes for exactly one of the two candidates, and the candidate with a majority wins, (a) define a sample space for the outcome of the election; (b) describe the event that Nolan receives at least three times as many votes as Ryan.

Solution: **(a)** Let x be the number of faculty members who vote for Ryan. Then $46 - x$ is the number of those who vote for Nolan. A sample space is

$$S = \{(x, 46 - x) : x = 0, 1, 2, \dots 46\}.$$

(b) The event that Nolan receives at least three times as many votes as Ryan is

$$E = \big\{(0, 46), (1, 45), (2, 44), \dots (11, 35)\big\}. \quad \blacklozenge$$

Example 1b Three numbers a, b, and c are selected randomly from the intervals $(-1, 1)$, $(0, 1)$, $(-1, 0)$, respectively. Define a sample space for this experiment and describe the event that the quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots.

Solution: Let $\mathbf{R}$ be the set of all real numbers. Then the sample space is

$$\big\{(a, b, c) \in \mathbf{R} \times \mathbf{R} \times \mathbf{R} \colon\ -1 < a < 1,\ 0 < b < 1,\ -1 < c < 0\big\},$$

and the desired event is $E = \big\{(a, b, c) \in S \colon b^2 > 4ac\big\}$, since the quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots if and only if its discriminant, $b^2 - 4ac$, is strictly positive. $\blacklozenge$

Example 1c Define a sample space for the experiment of flipping a coin 12 times, and describe the following events:

(a) the fifth flip is heads;

(b) at least one outcome is heads.

Solution: A sample space for this experiment is

$$S = \{\omega_1\omega_2\cdots\omega_{12} \colon \omega_i = \text{H or T}, \quad 1 \le i \le 12\}.$$

The event that the fifth flip is heads is

$$A = \{\omega_1\omega_2\omega_3\omega_4\text{H}\omega_6\cdots\omega_{12} \colon \omega_i = \text{H or T}, \quad 1 \le i \le 12, i \ne 5\}.$$

Let

$$A_1 = \{\text{H}\omega_2\omega_3\cdots\omega_{12} \colon \omega_i = \text{H or T}, \quad 2 \le i \le 12\};$$

for $2 \le j \le 11$, let

$$A_j = \{\omega_1\omega_2\cdots\omega_{j-1}\text{H}\omega_{j+1}\cdots\omega_{12} \colon \omega_i = \text{H or T}, \ 1 \le i \le 12, i \ne j\};$$

and let

$$A_{12} = \{\omega_1\omega_2\cdots\omega_{11}\text{H} \colon \omega_i = \text{H or T}, \ 1 \le i \le 11\}.$$

The event of at least one heads is $\bigcup_{j=1}^{12} A_j$. ♦

Example 1d A farmer decides to build a pen in the shape of a triangle for his chickens. He sends his son out to cut the lumber and the boy, without taking any thought as to the ultimate purpose, makes two cuts at two points selected at random. (a) Describe a sample space for this experiment. (b) Describe the event that the resulting three pieces of lumber can be used to form a triangular pen.

Solution: **(a)** Let x be the distance from the left end of the lumber to the leftmost break. Let y be the distance from the left end of the lumber to the rightmost break. The sample space is $S = \{(x, y) \colon 0 < x < y < \ell\}$.

(b) The three pieces x, $y - x$, and $\ell - y$ form a triangle if and only if

$$\begin{aligned} x &< (y - x) + (\ell - y) \\ (y - x) &< x + (\ell - y) \\ \ell - y &< x + (y - x), \end{aligned}$$

or, equivalently,

$$x < \frac{\ell}{2}, \quad y < x + \frac{\ell}{2}, \quad \text{and} \quad y > \frac{\ell}{2}.$$

So the event that the resulting three pieces of lumber can be used to form a triangular pen is $E = \left\{(x, y) \colon 0 < x < \dfrac{\ell}{2} < y < x + \dfrac{\ell}{2}\right\}$. ♦

Example 1e Travis picks up darts to shoot toward an $18''$-diameter dartboard aiming at the bullseye on the board. Consider a rectangular coordinate system with origin at the center of the dartboard. The face of the dartboard is divided into 20 equal sectors (wedged-shaped slices) with the positive x-axis being a straight side of the first sector (see Figure 1a). Define a sample space for the point at which a dart hits the board. For this sample space, describe the event that the dart hits a sector that has the positive y-axis as a side.

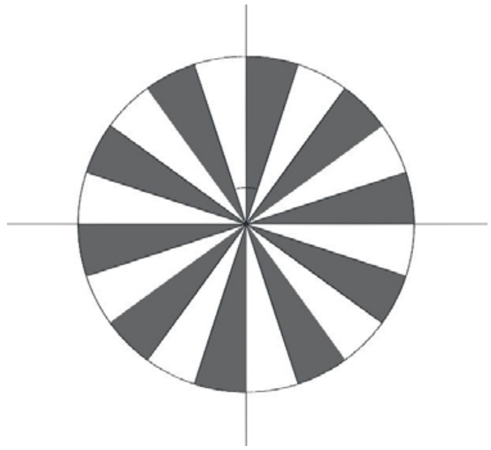

Figure 1a Dart Board of Example 1e.

Solution: An appropriate way to define a sample space for this experiment is to use polar coordinates:

$$S = \{(r, \theta)\colon 0 \le r < 9,\ \ 0 \le \theta < 2\pi\}.$$

The angle of each sector of the board is $2\pi/20 = \pi/10$. The fifth and sixth sectors of the board are the only sectors that have the positive y-axis as one of their straight sides. So the desired event is

$$E = \Big\{(r, \theta)\colon 0 \le r < 9,\ \ \frac{4\pi}{10} \le \theta \le \frac{6\pi}{10}\Big\} = \Big\{(r, \theta)\colon 0 \le r < 9,\ \ \frac{2\pi}{5} \le \theta \le \frac{3\pi}{5}\Big\}. \quad \blacklozenge$$

Example 1f When cell phones were not common, in almost all towns, public pay phones were installed on the sidewalks. Once upon a time, in Berkeley, California, one such a phone was not working properly. It returned coin deposits 50% of the time whether or not the calls went through, it connected 30% of the outgoing calls to wrong numbers, and 23% of the time it took the coin but connected the call to an incorrect number. What is the probability that a call made by this phone was connected for free to the right number?

Solution: Let A be the event that the coin deposited was returned, and B be the event that the call made was connected to the right number. We are interested in $P(AB)$. We have

$$\begin{aligned} P(AB) &= P(A) + P(B) - P\big(A \cup B\big) \\ &= P(A) + P(B) - \big[1 - P(A^c B^c)\big] \\ &= 0.50 + 0.70 - (1 - 0.23) = 0.43. \quad \blacklozenge \end{aligned}$$

Example 1g Show that for three events A, B, and C, the probability that exactly one of them occurs is

$$P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC).$$

Solution: The event that exactly one of the three events A, B, or C occurs is

$$A \cup B \cup C - \big(AB \cup AC \cup BC\big).$$

Since $AB \cup AC \cup BC \subseteq A \cup B \cup C$, we have that the desired probability is

$$\begin{aligned}
&P\big(A \cup B \cup C\big) - P\big(AB \cup AC \cup BC\big) = \\
&\quad P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC) - \\
&\quad \big[P(AB) + P(AC) + P(BC) - P(ABC) - P(ABC) - P(ABC) + P(ABC)\big] = \\
&\quad P(A) + P(B) + P(C) - 2P(AB) - 2P(AC) - 2P(BC) + 3P(ABC). \quad ♦
\end{aligned}$$

Example 1h In Italy, buses that travel between Rome and Palermo usually stop in a certain village for lunch. Let E, F, and H be the events that a randomly selected entrée from the restaurants of this village is salty, spicy hot, and fatty, respectively. Suppose that $P(E) = 0.08$, $P(F) = 0.15$, $P(H) = 0.17$, $P\big(E \cup F\big) = 0.19$, $P\big(F \cup H\big) = 0.25$, and $P(EFH) = 0.02$. Lucrezia orders an entrée from a restaurant in this village randomly. Find the probability that (a) it is not salty; (b) it is salty and spicy hot; (c) it is spicy hot and fatty, but not salty.

Solution: (a) $P(E^c) = 1 - 0.08 = 0.92$;
(b) $P(EF) = P(E) + P(F) - P\big(E \cup F\big) = 0.08 + 0.15 - 0.19 = 0.04$.
(c) To find the desired probability, $P(E^cFH)$, note that $E^cFH = FH - EFH$, and $EFH \subset FH$. Thus

$$P(E^cFH) = P(FH) - P(EFH).$$

But

$$P(FH) = P(F) + P(H) - P\big(F \cup H\big) = 0.15 + 0.17 - 0.25 = 0.07.$$

Therefore, the desired probability is $P(E^cFH) = 0.07 - 0.02 = 0.05$. ♦

Example 1i For events E, F, and G show that

(a) $P(EF) \geq P(E) - P(F^c)$.

(b) $P(EFG) \geq 1 - P(E^c) - P(F^c) - P(G^c)$.

Solution:

(a) By Theorem 1.7 of the book, we have that $P(E) = P(EF) + P(EF^c)$. So

$$P(EF) = P(E) - P(EF^c).$$

Now $EF^c \subseteq F^c$, so $P(EF^c) \leq P(F^c)$. Hence

$$P(EF) = P(E) - P(EF^c) \geq P(E) - P(F^c).$$

(b) By part (a)

$$P(EFG) \geq P(EF) - P(G^c) \geq P(E) - P(F^c) - P(G^c)$$
$$= 1 - P(E^c) - P(F^c) - P(G^c). \quad \blacklozenge$$

Example 1j A coin is tossed infinitely many times. Is the sample space of this experiment countably infinite or uncountably infinite?

Solution: In a toss of the coin, let 1 denote an outcome of heads and 0 denote an outcome of tails. The sample space of the experiment of tossing a coin indefinitely is the set of all sequences of 0's and 1's. Clearly, there is a one-to-one correspondence between such a sequence and the numbers of the interval $[0,1]$ expressed in the binary numeral system; i.e., expressed in the base-2 numeral system. Since $[0,1]$ is an uncountable set of points, the experiment of tossing a coin infinitely many times has a sample space that is uncountably infinite. ♦

Example 1k A point is chosen at random from the interval $(-1,1)$. Let E_1 be the event that it falls in the interval $(-1,1/3]$, E_2 be the event that it falls in the interval $(-1,1/9]$, E_3 be the event that it falls in the interval $(-1,1/27]$ and, in general, for $1 \leq i < \infty$, E_i be the event that the point is in the interval $(-1,1/3^{\,i}]$. Find $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$.

Solution: Clearly, $E_1 \supset E_2 \supset E_3 \supset \cdots \supset E_i \supset \cdots$. Hence $\bigcup_{i=1}^{\infty} E_i = E_1 = (-1,1/3]$. Now all the points of the interval $(-1,0]$ belong to all E_i's, and for any other point, x, $x \in (0,1)$, there is an i for which $x \notin (-1,1/3^i)$. So $\bigcap_{i=1}^{\infty} E_i = (-1,0]$. ♦

In the following section, no prior knowledge of genetics is assumed. If it is skipped, then all exercises and examples in this and future companions for various chapters marked "(Genetics)" should be skipped as well.

1B APPLICATIONS OF PROBABILITY TO GENETICS

In 1854, the Austrian monk Greger Johann Mendel (1822–1884) proposed a mechanism for the inheritance of physical traits. Since then probability has played a central role in the understanding of certain aspects of genetics. In this section, we will examine some elementary applications of probability in the foundations of modern genetics.

Biologists estimate that the human body contains approximately 60 trillion cells. The width of a cell is approximately one-hundredth of a millimeter. Within each cell, there is a large oval structure called the **nucleus**. In the nucleus, the control center of the cell, are threadlike bodies called **chromosomes**. Chromosomes contain genes. In the human, there are about 100,000 genes. A **gene** is the fundamental physical unit of heredity. Genes code for characteristics of the individual and are passed on from one generation to another. For example, a person with green eyes has genes that code for green eyes, and a person with a bent little finger has genes that code for that.

Human beings have 46 chromosomes (23 pairs) in each cell. This number is constant for a particular species. For example, chickens have 36 and peas have 14 chromosomes in each of their cells, respectively. In humans, the reproductive cells are the egg and sperm cells. Except for the reproductive cells, in each human cell, 23 chromosomes come from the father and 23 from the mother. Each member of a pair, is the same length and has the same appearance. It is important to know that the genes that control a certain characteristic occupy the same location on the paired chromosomes. Of all the genetic material on the chromosomes, there are between 20,000 and 50,000 genes that code for all characteristics of the individual. For nonreproductive cells, every cell carries every single gene. Therefore, all of the genetic material of individuals can be found in each and every cell. In each nonreproductive cell, there is one pair of chromosomes called the *sex chromosome*, which determines a person's gender. Unlike all other pairs of chromosomes, in males, the sex chromosomes do not look alike. One is called the X chromosome and one is called the Y. They differ in shape and size. Females have two X chromosomes. Furthermore, in males, the X chromosome has no corresponding genetic information on the Y chromosome. The reproductive cells contain *only* 23 unpaired chromosomes. This is so that during fertilization each parent can contribute 23 chromosomes to the zygote or fertilized egg. We will discuss sex chromosomes and reproductive cells in Companion for Chapter 3. For now, we concentrate on nonreproductive cells and nonsex chromosomes.

Genes appear in alternate forms. For example, for seed color of the garden pea, the gene appears in one form for yellow and another form for green. Similarly, the gene for the stem height of the garden pea has one form for "tall" and another form for "short." Alternate forms of genes are called **alleles**. Therefore, the gene for the seed color of the garden pea has two alleles, the allele for yellow and the allele for green. The gene for the stem height of a garden pea has an allele for tall and an allele for short.

During fertilization, for each trait, each parent contributes only one of a pair of alleles. That is how the paired conditions of chromosomes and genes, discussed previously, are transferred from one generation to another. The genes contributed by the parents may or may not be the same allele. For the seed color of the garden pea, during reproduction, both parents might contribute their yellow producing alleles, both might contribute their green producing alleles, or one might contribute its green and the other its yellow producing allele. In the former case, after reproduction, it does not matter which parent contributed the yellow producing allele and which the green producing allele. The result is the same: The offspring has one yellow and one green producing allele. It often happens that when the genes contributed by the parents are different alleles, then, in the offspring, only one allele is fully expressed and the other one is masked. That is, one allele is **dominant** over the other. In such cases, one can observe the full effect of the dominant allele, but no effect of other alleles can be detected. The allele that is masked is called **recessive**.

As Mendel did, suppose that a biologist takes a number of pea plants that have inherited one dominant allele for purple flower color from father and one dominant allele for purple flower color from mother. Suppose that the biologist crosses each of these plants with a pea plant that has inherited two recessive alleles for white flower color, one from each parent. Then she will observe that the resulting first generation offspring all will have purple flowers and, as a result of fertilization, the colors will not change to lighter purple. This is an observation that proves that inheritance is not *necessarily* a blending phenomenon. At this stage, one might think that the absence of white flowers among the first generation offspring indicates that its heritable factor is annihilated. However, if the biologist continues her experiment and crosses the new generation of the plants between each other, she will observe that approximately 1/4th of their offspring will have white flowers. This phenomenon demonstrates the presence of the recessive white flower allele in the first generation of the plants.

The organism's genetic constitution (makeup); that is, the set of genes that it possesses, is its **genotype**. Thus, if in an organism, an inheritable characteristic is controlled by a single pair of genes, one from each parent, then the pair is the genotype of that characteristic. To discuss applications of probability to genetics, we use uppercase letters to represent dominant alleles and lowercase to represent recessive alleles. For the pea plants, let A represent the dominant allele for purple flowers and a represent the recessive allele for white flowers. Then each of the pea plants in the experiment above belongs to one of the three genotypes AA, Aa, aa. Recall that Aa and aA are identical. They both mean that the pea plant has inherited a dominant purple flower allele from one parent and a recessive white flower allele from another parent. AA and aa are defined similarly. For Mendel's experiments with pea plants, to determine all the possible genotypes of offsprings, the possible results for the first set of experiments is conveniently represented in the following table:

	A	A
a	aA	aA
a	aA	aA

This table shows that the only outcome is aA, and since A is dominant, the probability that all offsprings have only purple flower is 1. In the next stage, these offsprings are crossed with each other. That is, pea plants of genotype aA are crossed with pea plants of genotype aA. The possible results are given by the following table.

	a	A
a	aa	aA
A	Aa	AA

Since all four genotypes in the center of this table are equally likely and aA and Aa are the same, the probability of genotypes aa, aA, and AA are 1/4, 1/2, and 1/4, respectively. This shows that the probability is 1/4 that a second generation offspring has white flowers.

Example 1ℓ Suppose that lima bean plants with green pods were crossed with lima bean plants with yellow pods. Furthermore, suppose that the parents of all the plants crossed came from pure lines. Therefore, each plant crossed had identical alleles. If all these crossings yielded 420 offsprings of which 100 had yellow pods and 320 had green pods, which allele is dominant, the allele for green pods or the allele for yellow pods?

Solution: Intuitively, it should be clear that the allele for a green pod is dominant. To show this mathematically, let A represent the dominant allele and a represent the recessive allele. Since the parents of all the plants crossed came from pure lines, as the following table shows, each lima bean plant crossed belonged to the genotype aA.

	A	A
a	aA	aA
a	aA	aA

Therefore, by the following table, each of the 420 offsprings belongs to one of the genotypes aa, aA, and AA.

	a	A
a	aa	aA
A	Aa	AA

Since a is the recessive allele, by this table, with probability 1/4, an offspring is a lima bean that has inherited two recessive alleles. Therefore, approximately 1/4th of the 420 offsprings; that is, approximately 105 of the plants should be lima bean plants that inherited two recessive alleles for their pod color. Since, in the experience, of the 420 offsprings, 100 had yellow pods and 320 had green pods, and 100 is statistically close enough to 105, the recessive allele is the allele for yellow pod, and hence the dominant one is the allele for green pod.

Remark In this exercise, the fact that, approximately 1/4th of the 420 offsprings should be lima bean plants that inherited two recessive alleles for their pod color should be intuitively clear. However, it is mathematically justified by the formula for the expected value of binomial random variables, which is discussed in Section 5.1 of the book. ♦

Example 1m In human beings, eye colors and existence or nonexistence of dimples are physical attributes that have an influence on our perception of facial beauty. As for eye colors, brown eyes (B) are dominant over both green eyes (G) and blue eyes (b). However, whenever there is no allele for brown eyes, but there are alleles for green and blue eyes, green is dominant over blue. When it comes to dimples, people with dimples have dominant alleles (D) over those without dimples who have recessive alleles (d). In this example, we are interested in two characteristics transferred from one generation to another. Namely, eye color and presence or absence of dimples. Suppose that a man with genotype $Bbdd$ marries a woman with genotype $bbDd$. Find the probability that a child of theirs (a) has dimples and brown eyes; (b) has no dimples, but brown eyes; (c) has blue eyes.

Solution: The child inherits Bd or bd from the father with equal probabilities, and bD or bd from the mother also with equal probabilities. So the child belongs to one of the four genotypes in the following table with equal probabilities.

	Bd	bd
bD	$BbDd$	$bbDd$
bd	$Bbdd$	$bbdd$

This table shows that the child has (a) dimples and brown eyes with probability 1/4; (b) no dimples and brown eyes with probability 1/4; and (c) blue eyes with probability 1/2. ⧫

EXERCISES FOR SECTION 1B

1. For the shape of a pea seed, the allele for a round shape, denoted by R, is dominant to the allele for a wrinkled shape, denoted by r. A pea plant with a round seed shape is crossed with a pea plant with a wrinkled seed shape. If half of the offspring have round seed shapes and half have wrinkled seed shapes, what were the genotypes of the parents?

2. In a study, 200 children of the parents who all have an allele for a cleft chin and an allele for a smooth chin are considered. If only 48 of them have a smooth chin, show that the gene for a cleft chin must be dominant over the gene for a smooth chin.

3. In human beings, the gene for naturally curly hair (H) is dominant over the gene for straight hair (h), and the gene for right-handedness (R) is dominant over the gene for left-handedness (r). Suppose that a man with genotype $Hhrr$ marries a woman with genotype $hhRr$. Find the probability that a child of theirs (a) is right-handed with curly hair; (b) is right-handed; (c) has straight hair.

4. In human beings, unattached ear lobes (E) are dominant over ear lobes that are attached to the sides of the head (e). The ability to roll the lateral edge of the tongue together (T) is a dominant trait over the inability to do so (t). As for dimples, people with dimples have dominant alleles D over those without dimples having recessive alleles d. Suppose that a man with genotype $EettDd$ marries a woman with genotype $Eettdd$. Find the probability that a child of theirs has dimples and unattached ear lobes.

5. A hitchhiker's thumb is a thumb that can bend backwards when extended. In human beings, no-hitchhiker's thumb (H) is dominant over hitchhiker's thumb (h), face freckles (F) are dominant over no face freckles (f), and brown eye color (B)

is dominant over blue eye color (b). Suppose that a man with genotype $HhffBb$ marries a woman with genotype $HhFfbb$. Find the probability that a child of theirs has (a) blue eyes and hitchhiker's thumb; (b) blue eyes and no-hitchhiker's thumb; (c) blue eyes, hitchhiker's thumb, and face freckles.

■

Answers to Odd-Numbered Exercises of Section 1B

1. The genotypes of the parents were rr and Rr.
3. The answers to parts (a), (b), and (c) are 1/4, 1/2, and 1/2, respectively.
5. The answers to parts (a), (b), and (c) are 1/8, 3/8, and 3/16, respectively.